

E-commerce Order Management

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Time limit: 48 hours from assignment. **Stack:** Spring Boot.

The Project

An e-commerce order management system at scale with a multi-category product catalog, customer cart and checkout, inventory tracked across multiple warehouses, payment processing, and an order fulfillment lifecycle (placed → confirmed → packed → shipped → delivered → returned). Inventory must be reserved correctly under concurrent purchases so the same unit cannot be oversold across warehouses. Order placement should atomically reflect cart, inventory, and payment state, and the downstream pipeline (fulfillment routing, customer notifications, audit logging) should run without blocking the customer's checkout response. Discounts, taxes, returns, and refunds should be supported.

Roles: admin (manage catalog, warehouses, inventory, and discounts), customer (browse, cart, checkout, return, and track orders), and warehouse staff (update fulfillment status).

Instructions

Framing

The Product Requirement above is intentionally open-ended. You own the scoping decisions — which entities to model, which APIs to expose, which edge cases to handle, and what to leave out. Your interpretation of the requirement and the features you choose to build are themselves part of what is being evaluated. You are encouraged to interpret the requirement generously and submit a feature-rich solution — both the breadth and the depth of the features you build contribute to the evaluation. Document every meaningful assumption in the `README.md` and explain your reasoning in the recorded video.

In Scope

The following are expected in your submission:

- REST APIs covering the core flows
- Persistence to a database of your choice
- Basic role-based access control for the roles defined in the requirement
- Input validation and error handling

- Unit & Integration Tests for the core flows

Out of Scope

Do not spend time on:

- UI or frontend
 - Deployment, containerization, or CI/CD
 - Distributed systems or microservices
 - Advanced authentication (OAuth, SSO, MFA)
 - Production-grade observability, monitoring, or alerting
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What to Submit

- **GitHub repository** (mandatory)
 - Your personal Github Project repository link
 - Multiple commits are expected during the development phase
 - Must include a `README.md`
 - Must include the `Agents.md` / `Claude.md` file used during development
 - Must include the skills used during development
 - Must include all raw files used during development
- **Video recording** (maximum 10 minutes [loom video](#)) in which you explain:
 - How you approached the problem and the solution at a high level
 - The tech stack used and the reasoning behind it
 - The AI workflow used
 - The testing approach